

NIAID-BRCs AI Codeathon 2.0

# Sample Metadata That Survives Analysis in Galaxy

Making metadata construction, validation, and Galaxy input mapping automatic — so the most naive user follows a good path that reduces risk and makes uncertainty transparent.

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# The Problem

Attempting an analysis with incomplete, poorly structured, dispersed, or absent sample metadata can cause **confident but subtly wrong** results with unpredictable consequences and magnitude of impact.

## What goes wrong

- Sample metadata is often missing, incomplete, or not structured in a way that maps to Galaxy data types
- The agent can help — but it makes confident errors: misreading papers, propagating upstream label errors, presenting inferred values as fact
- Users don't know to ask the right questions, and the agent doesn't volunteer what it's uncertain about
- Errors propagate silently into the analysis, where they become hard to distinguish from real biological signal

## Where We Work: Orbit

**Orbit** is an AI-powered desktop environment for biological analysis, built on **Loom** (the agent brain) and **Galaxy** (the execution engine).

- **Loom** is the brain — it plans, reasons, and drives the notebook
- **Galaxy** is the primary execution path — workflows, tools, reproducible runs
- **Skills** are modular instructions the agent fetches on demand, driven by frontmatter that tells it *when* to use them
- The **notebook** is the durable record — plans, decisions, results, and evidence persist there

Skills fire automatically based on what the user asks — the user doesn't need to know they exist.

## Case Study: A Known Quantity

We asked the agent to reconstruct sample metadata from public resources for a **published PNAS paper from a team member** — so we could check its output against ground truth.

### The paper

Romano et al., *PNAS* 2024 — "Perturbation-specific transcriptional mapping for unbiased target elucidation of antibiotics"

GEO GSE251671 · 1,059 samples · SRA PRJNA1055047 · 1,310 runs · *P. aeruginosa* PA14

The agent was **mostly able** to reconstruct the metadata. It located the study, pulled sample attributes, and assembled a structured record without much prompting.

But the sample sheet it produced **required archeology from someone with intimate knowledge** of the study to verify — and it was subtly and confidently wrong in two places.

## Two Confident Errors

### Error 1: Water dosed

#### Negative control dosage misread

The agent recorded a dosage of ~0.05 for the negative control, which was just **water**. Water is not dosed. This was a misread of the paper, presented as fact without flagging it as uncertain.

### Error 2: Single vs. paired

#### Upstream mislabeling propagated

The agent labeled many samples as "single" when they were actually paired. This was an upstream SRA error — but the allegedly single samples each had **two files with the same number of spots**, strong evidence of paired-end data. The agent propagated the label without questioning it against the data.

Both errors are the kind a careful human analyst would catch. The agent didn't flag them because it had no instruction to check inferred values against the data or surface uncertainty.

## Learning From the Process

We asked the agent to do this assessment. **The average user won't** — they'll just start talking about analyzing data and expect the agent to handle the rest.

Two goals emerged:

### Goal 1

#### Make the process honest

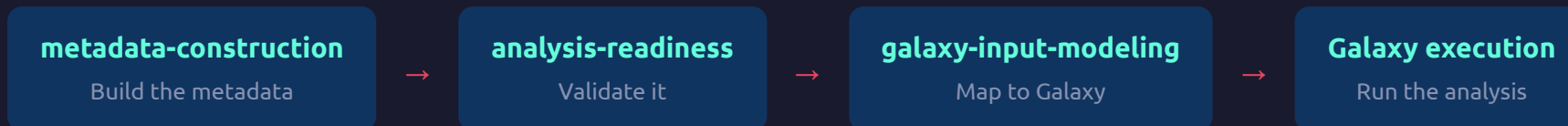
Flag inferred values, conflicts, and unresolved questions to the user rather than presenting them as fact.

### Goal 2

#### Make it automatic

Metadata construction and validation should kick in when a user starts talking about analyzing data — without knowing these steps exist.

## Three Skills, In Order



Each skill is self-contained, discovered by Orbit via frontmatter, and fires automatically based on what the user asks. The user never needs to know they exist.

The chain has a **hard gate**: analysis does not proceed until readiness declares the study ready or ready with warnings.

# What Each Skill Does

## 1. metadata-construction

### Build structured metadata from all available sources

- **Public data:** track down metadata from NCBI, EBI, GEO, linked publications *before* asking the user
- **User's own data:** interview them for the experimental design
- Every value gets provenance: `reported`, `observed`, `inferred`, `conflicting`, or `unknown`
- Produces a provenance YAML + a tabular sample sheet

## 2. analysis-readiness

### Validate against the data and the intended analysis

- Open-ended scope — anything that could undermine validity, not a fixed checklist
- Tests claims against the actual data (read counts, file contents, URL resolution)
- Classifies issues as blocking, warning, or informational
- Declares `ready`, `ready_with_warnings`, or `blocked`

## 3. galaxy-input-modeling

### Convert validated metadata to the right Galaxy structure

- Inspects the live tool/workflow input contract — never guesses from memory
- Chooses: dataset, list, paired, list:paired, nested, or sample sheet
- Verifies collection shape and contents, not just job completion

## With Skills: What Changes

Re-running the Broad case study with the skills enabled. *(Results pending — this run has not yet been completed.)*

Instead of silently filling gaps, the construction skill records **where every value came from** and **how confident it is**:

### Water control — uncertainty flagged

#### Negative control dosage

```
sample_id: GSM7985465
condition: water
pert_dose:
  value: 0.05
  provenance: inferred
  source: >
    Paper: "compounds in 0.5% (v/v)
    water (or DMSO) in Lysogeny
    Broth"
  confidence: low
  notes: >
    0.5% is the vehicle volume, not
    a dose. Water is the negative
    control. Flagged to user: this
    value does not make sense for
    a water control.
```

### Library layout — conflict flagged

#### Single vs. paired

```
sample_id: GSM7985294
layout:
  reported: paired
    source: paper (methods: "paired-
    end sequencing")
  observed: single
    source: SRA run data
  provenance: conflicting
  notes: >
    Paper says all paired-end but
    SRA run data says single for
    this sample. Flagged to user.
```

# A Self-Refining System

The skills are a prototype — essentially suggestions the agent follows. The system should also **learn from its own failures** and feed those lessons back.

Agent encounters a failure mode



Pushes finding to shared DB



Harvested to improve skills

## Where the lessons go

- **Improve future skills** — failure patterns become new checks
- **Improve the harness itself** — the most impactful suggestions become rules in Orbit, not just skills
- **Inform upstream submission** — systemic issues (like SRA mislabeling) can be flagged to NCBI at data submission time
- Uses a similar mechanism to user feedback — the agent already has a feedback channel

## Future Directions

### Eval suite

#### Measure what works

A formal set of studies with known ground truth, run with and without the skills, scored on metadata accuracy, gap detection, and whether inferred values are honestly labeled. This lets us quantify improvement and identify which checks make the greatest difference.

### Skills → harness rules

#### Enforce, don't just suggest

The skills that prove most impactful — based on the eval suite — migrate from suggestions into rules in the Orbit harness itself, so they are enforced rather than optional.

### Upstream feedback

#### Fix it at the source

Self-feedback could identify systemic issues upstream (e.g. SRA mislabeling) and pass those along at data submission time. Lessons learned could be repurposed to prepare data for submission so future submissions start more complete.